

ASIA ECONOMICS ANALYST

Food inflation in ASEAN – When oil shock meets El Niño

Key Messages:

- **Regional food prices face upside risk in coming months from three shocks:** oil, fertilizer and climate conditions. First, the oil shock from the Middle East conflict has shown up in fuel-sensitive CPI items, and higher fertilizer prices will raise farm input costs. A potential strong El Niño event in late 2026 could create another food supply shock just as oil and fertilizer pressures are passing through the food chain.
- **ASEAN is exposed through trade, CPI weights and commodity linkages.** Food inflation is highly synchronized across the region and food carries a large weight in CPI baskets. Singapore and the Philippines are net food importers, while Malaysia's and Indonesia's food surpluses are concentrated in palm oil. Even Thailand, the region's net food exporter, relies heavily on imported fertilizer.
- We estimate **a 10% increase in local oil prices raises ASEAN food CPI by around 0.3pp after 12 months**, all else equal. Fertilizer effects are smaller at 0.2pp, while El Niño effects are less precisely estimated.
- **The forecast path points to additional pipeline pressures.** Feeding our commodities team's forecast and NOAA's El Niño probability-weighted projections through our model implies additional ASEAN food inflation of around 2.1pp, on average, after 12 months. Indonesia, Philippines and Thailand face the clearest upside risks, while the impact on Singapore and Malaysia looks more contained.
- **Policy buffers can blunt El Niño pass-through.** Past episodes show that food inventory releases, import liberalization, subsidies and price controls have often cushioned domestic food prices.

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A potential food inflation shock ahead

The conflict in the Middle East has driven crude oil prices sharply higher, while the initial blockade of key shipping lanes has squeezed fertilizer supply. The first-round inflation impact is straightforward: higher oil prices show up first in fuel-sensitive components such as transport and airfare. However, the second-round inflation impact, which comes with a lag, is likely to start feeding into the food-price pipeline.

An additional complication is weather. The National Oceanic and Atmospheric Administration (NOAA) has assessed that El Niño conditions are present, with forecast models pointing to a meaningful risk of a very strong event around November 2026 to January 2027. If realized, this would add another supply shock just as higher oil and fertilizer costs are still passing through the food chain.

ASEAN is not uniformly exposed, but the region is vulnerable through several channels. The region includes near-total food importers as well as major rice and palm oil producers, while food carries a large weight in CPI baskets. In this report, we quantify the potential impact on ASEAN food prices from three shocks: oil, fertilizer and El Niño.

ASEAN looks particularly vulnerable

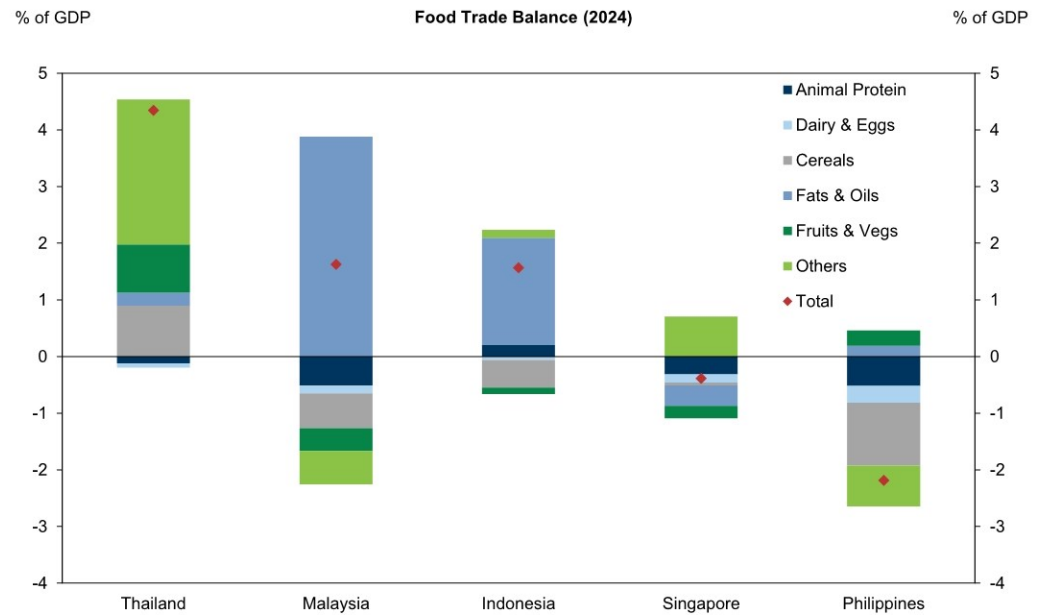
ASEAN food inflation is highly synchronized. A simple principal component analysis shows that the first common factor explains around 70% of the variation in food inflation across the region, underscoring how regional food prices tend to move together during major shocks.

Singapore and the Philippines are net food importers, leaving them directly exposed to global food price shocks. Malaysia and Indonesia appear more insulated in aggregate, but their food surpluses are largely a palm-oil story; excluding that component, both become net food importers (Exhibit 1). Thailand is the only broad-based net food exporter in the region. Even then, Thailand is still exposed to global food price shocks through increases in food input prices, such as fertilizer, more than 90% of which Thailand imports. The region's main food commodity imports are cereals, meat and dairy products.

The unprecedented oil shock is also forcing governments to reconsider the food-versus-fuel trade-off. To reduce reliance on imported fuel, Indonesia, Malaysia and Thailand have accelerated higher biodiesel blends, pulling more palm oil into energy use (Exhibit 2). That adds a new demand shock to a market already constrained by yields, while El Niño-related rainfall shortfalls could further pressure palm oil supply. Higher oil prices therefore risk tightening edible-oil markets as well as transport costs, reinforcing why ASEAN looks particularly vulnerable to a broader food inflation shock.

Exhibit 1: ASEAN economies are largely net food importers

Excluding palm oil, Malaysia and Indonesia are net food importers



Source: ITC Trade Map, Goldman Sachs Global Investment Research

Exhibit 2: The latest biofuel mandates in ASEAN

Country	Policy
Indonesia	From 1 July, all diesel fuel must be 50% palm oil, up from 40%; gasoline in Jakarta and other areas of Java island must be 5% ethanol
Malaysia	From 1 June, diesel suppliers should provide 15% palm oil blend, up from 10%
Philippines	Government council pushing for raising required blend of coconut oil in diesel fuel to 5% from 3%
Thailand	From 14 March, regular diesel fuel must be 7% palm oil, up from 5%; supplies of 20% blend increased for large vehicles; exports of crude palm oil restricted from 7 April

Source: Asia Nikkei, Goldman Sachs Global Investment Research

How do supply shocks affect food inflation in ASEAN?

We use the local projections (LPs) method of [Jordà \(2005\)](#)¹ to quantify the impact of oil, fertilizer and El Niño shocks on ASEAN food inflation. LPs are well suited to this exercise because they estimate the response of food CPI directly at each future horizon, without requiring us to specify the full underlying multivariate system.

Following [Borrillo et al. \(2024\)](#)², our outcome variable is each country's food CPI, while the key predictors are oil prices, fertilizer prices, El Niño conditions and industrial production as a proxy for economic activity. We adapt their framework in three ways for ASEAN. First, we use the Relative Oceanic Nino Index (RONI)³ rather than the standard

¹ Estimation and Inference of Impulse Responses by Local Projections.

² "El Niño" and "La Niña": Revisiting the impact on food commodity prices and euro area consumer prices.

³ In February 2026, NOAA officially adopted the Relative Oceanic Nino Index (RONI) for monitoring sea surface temperatures and forecasting El Niño and La Niña events. Whereas the traditional Oceanic Nino Index (ONI) uses a static 30-year period of recorded ocean temperatures to calculate departures from average, RONI evolves from month to month, allowing the index to better identify El Nino and La Nina events.

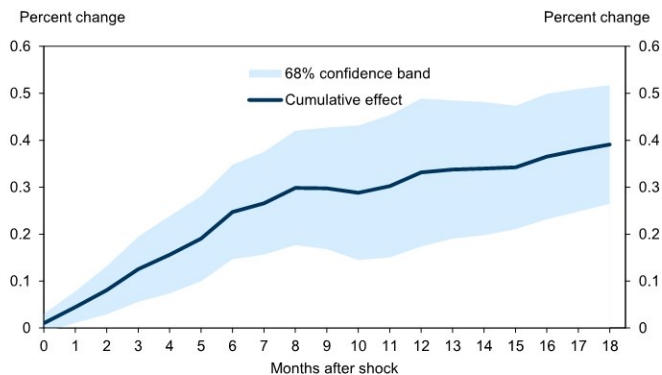
Oceanic Nino Index (ONI) to better capture ENSO (El Niño–Southern Oscillation) conditions relative to recent warming trends. Second, we measure the El Niño shock as the month-on-month change in RONI, while controlling for the lagged RONI level to capture the prevailing El Niño state. Third, we estimate both a pooled ASEAN panel LP and country-level LPs to distinguish the average regional response from country-specific heterogeneity. Given data constraints, the estimation sample starts from 2002.

On average, oil shocks generate the clearest pass-through to food prices: a 10% increase in local-currency oil prices raises food inflation by around 0.3% after 12 months, with statistically significant effects across the report horizon ([Exhibit 3](#)). Fertilizer effects are smaller but build over time, reaching 0.2% after 12 months ([Exhibit 4](#)). The average El Niño effect is less precisely estimated ([Exhibit 5](#)).

Country-level LP estimates point to material heterogeneity beyond the regional average. Oil pass-through is the most consistent result: it is positive in all countries and statistically significant in most. Fertilizer effects are generally smaller and less uniform, while El Niño responses vary more sharply across countries.

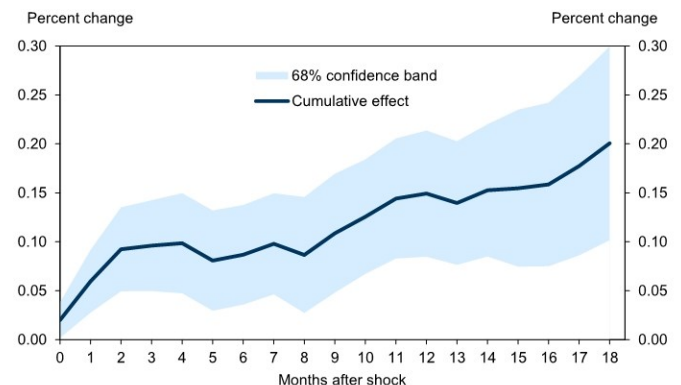
At the global level, our Commodities team [notes](#) that strong El Niño events have not typically been inflationary for global food prices. This is consistent with our more mixed El Niño estimates. We think domestic policy responses such as the release of buffer stocks, import liberalization, administered prices, and other policies blur the El Niño shock estimates.

Exhibit 3: Pooled ASEAN cumulative food CPI response (%) to a 10% increase in oil prices



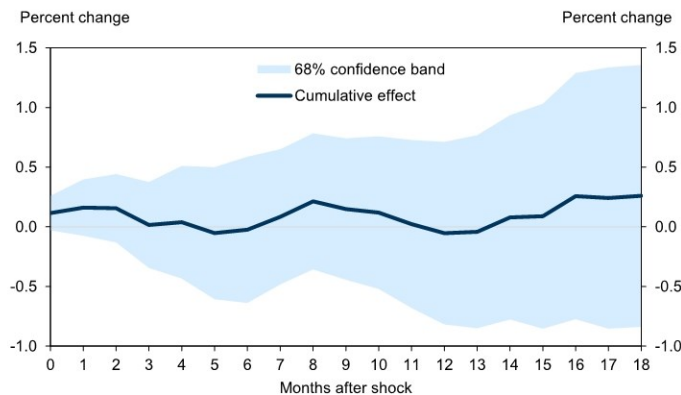
Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 4: Pooled ASEAN cumulative food CPI response (%) to a 10% increase in fertiliser prices



Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 5: Pooled ASEAN cumulative food CPI response (%) to a 1°C increase in El Niño (RONI)



Source: Haver Analytics, Goldman Sachs Global Investment Research

Pipeline pressure still building

Using the estimated responses over time after a shock, we translate the recent commodity shock and projected El Niño path into an additional impulse to food prices. The shock is already large: by end-May, Brent and urea prices were still around 46% and 63% above their February levels, even after easing from April's peak. At the same time, NOAA projects that RONI will rise from zero in May 2026 to around 1.7 degrees Celsius by late 2026, adding a potential weather shock on top of higher input costs.

We estimate that the oil, fertilizer and El Niño shocks could add, on average, 1.0pp to ASEAN's food inflation after six months and 2.1pp after 12 months, before moderating to 2.0pp in 18 months, relative to a no-shock baseline path. These estimates should be read as additional pressure on top of the usual food inflation trend, not as forecasts of total food inflation.

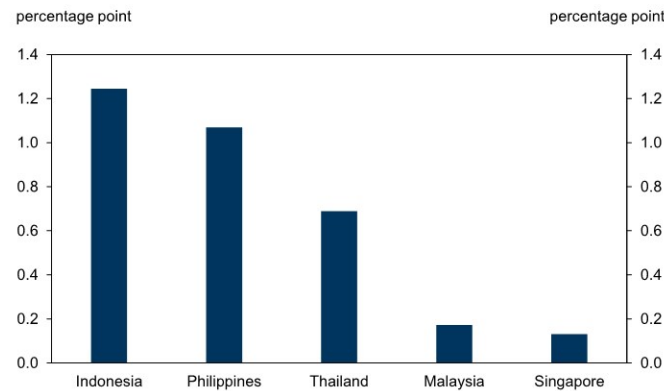
The average also masks wide dispersion: Indonesia, Philippines and Thailand face the clearest upside risks, while potential impacts on Singapore and Malaysia look more contained. In the following table, we show the cumulative percentage-point impact on headline CPI, calculated by multiplying the estimated food CPI impulse by each country's food weight in the CPI basket.

Exhibit 6: Percentage point contribution to headline CPI from oil, fertilizer and El Niño shocks

Country	Food weight in CPI	6m	12m	18m	Peak	Peak Month
Indonesia	22.5%	0.4	0.8	1.1	1.2	16
Philippines	34.8%	-0.1	0.3	1.1	1.1	18
Thailand	22.1%	0.3	0.7	0.2	0.7	13
Malaysia	15.6%	0.1	0.2	0.0	0.2	7
Singapore	6.5%	0.1	0.1	0.1	0.1	12

Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 7: Peak headline CPI response from oil, fertilizer and El Niño shocks



Source: Haver Analytics, Goldman Sachs Global Investment Research

Domestic policy responses

Around El Niño episodes, ASEAN governments have often introduced discretionary measures to cushion food prices. Some were not explicitly El Niño-specific but still overlapped with the shock and materially affected inflation through stock releases, import liberalization, subsidies or price controls. These interventions likely weakened the observed pass-through from El Niño to food prices, making our estimates less precise. Below, we highlight selected domestic policy responses, though the list is not exhaustive.

Release of buffer stock: In Thailand, the government's 2011-14 rice-pledging scheme, though fiscally costly, left it holding a record stockpile of roughly 17.8 million tons of rice. While the 2015-16 El Niño drought lowered rice production, the government's steady release of this stockpile actively held prices down, and the pass-through to consumer price inflation was modest. During the episode, Thailand also acted as a regional supply anchor, exporting its surplus to help keep neighbors such as the Philippines and Singapore supplied.

In Indonesia, the State Logistics Agency (Bulog) procures and holds national rice reserves, then releases stock when market prices come under pressure. This buffer proved useful after Indonesia's heavy rice imports in 2018, which reached 2.25 million tons worth over US\$1 billion. When drought cut rice output by 7% in 2019, Bulog's large carryover stocks helped cushion the shock and keep domestic rice prices broadly stable.

Import liberalization: In the Philippines, the 2019 Rice Tariffication Law was not an El Niño-specific measure, but it overlapped with the drought episode and helped cushion food price pressure by liberalizing rice imports. The law replaced quantitative import restrictions with tariffs, allowing much freer private-sector rice imports and expanded domestic supply. As imports rose, rice prices fell: the Philippines Department of Finance said rice tariffication helped cut rice prices in 2019, while the World Bank later described the liberalized sector as supporting stable food prices and cited 17 consecutive months of negative rice inflation after May 2019.

Subsidies/price ceilings: During the 2023/2024 El Niño episode, which coincided with India's rice export ban, the Malaysian government maintained the price cap on local

white rice despite rising input costs. This eventually triggered shortages that pushed demand toward imported rice and drove its price higher, prompting the government to extend subsidies to imported rice as well. Price controls stretched beyond rice: eggs and chicken were cushioned by subsidies and price ceilings, and sugar prices were held fixed through direct incentives paid to sugar producers.

Exhibit 8: Cumulative food CPI response (%) to a 10% increase in oil prices

Country	6m	12m	18m
Thailand	0.34**	0.56***	0.57***
Singapore	0.05	0.20	0.21
Malaysia	0.23*	0.33*	0.27
Indonesia	0.40*	0.32	0.64**
Philippines	0.20	0.32	0.55

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$

Source: Goldman Sachs Global Investment Research

Exhibit 9: Cumulative food CPI response (%) to a 10% increase in fertiliser prices

Country	6m	12m	18m
Thailand	0.10	0.24**	0.25*
Singapore	0.09*	0.16**	0.22**
Malaysia	0.10**	0.16**	0.22**
Indonesia	0.14	0.09	0.11
Philippines	-0.16	-0.20	-0.25

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$

Source: Goldman Sachs Global Investment Research

Exhibit 10: Cumulative food CPI response (%) to a 1°C increase in El Niño (RONI)

Country	6m	12m	18m
Thailand	-0.11	0.46	-0.06
Singapore	0.28	0.19	0.47
Malaysia	0.19	-0.27	-0.47
Indonesia	0.06	0.96	2.37
Philippines	1.95	2.95	3.07

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$

Source: Goldman Sachs Global Investment Research

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